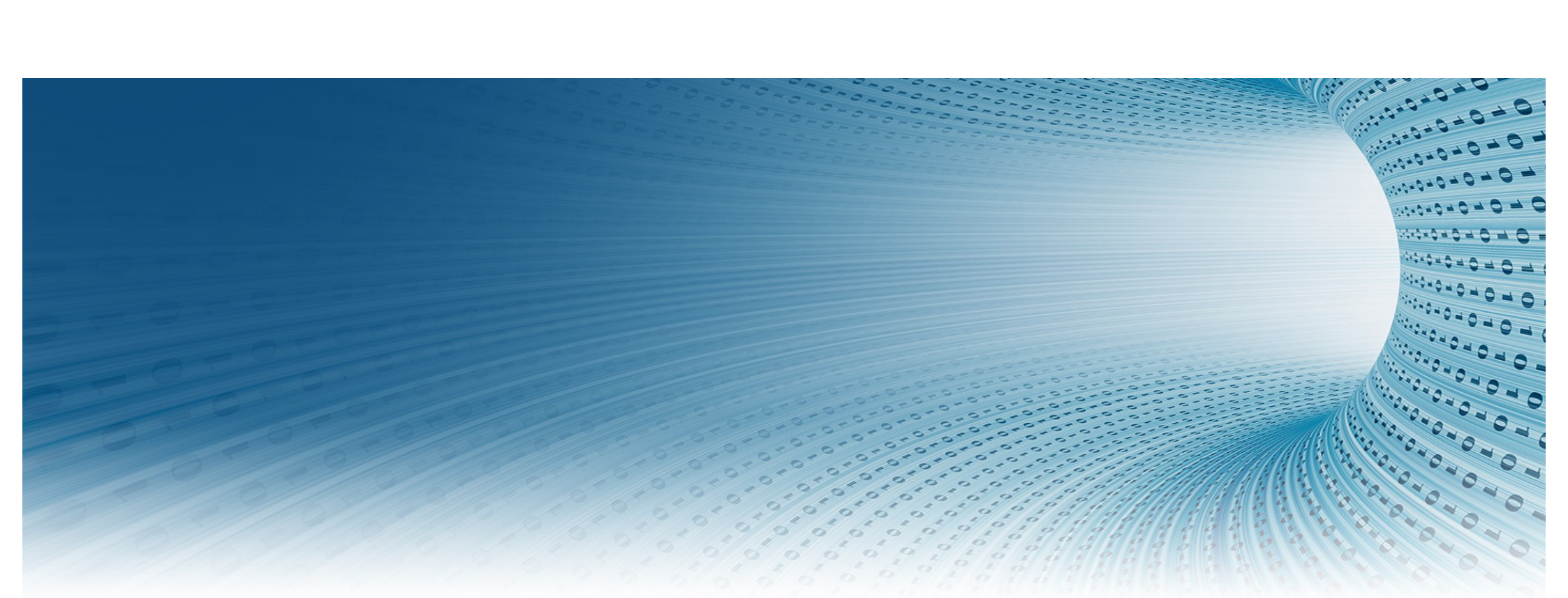


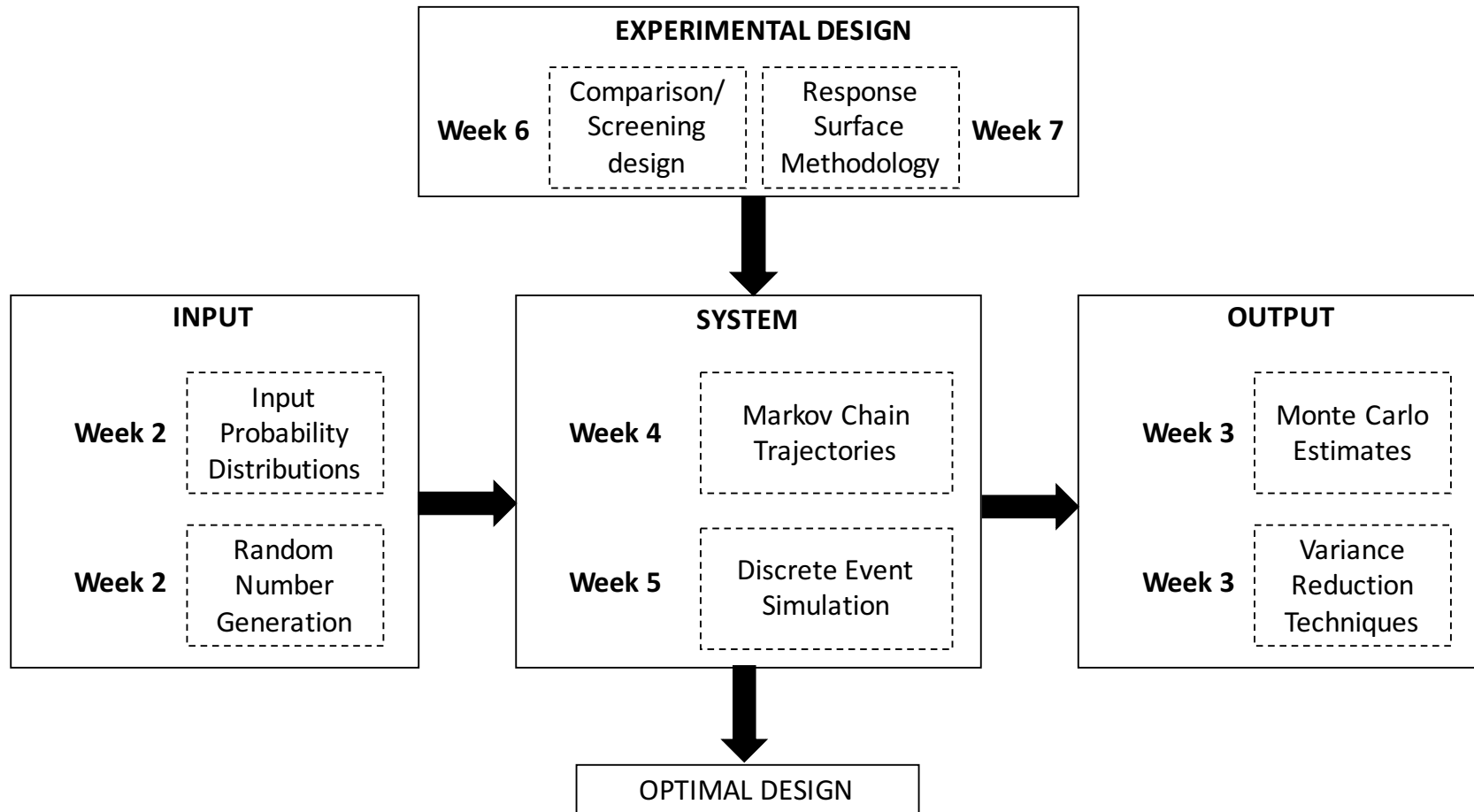


Experimental Design and Optimisation

Theory and formulation of Comparative designs

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Course Schedule



Outline

- Introduction
- Experimental design
- Comparison between design alternatives
 - Comparison between two design alternatives
 - Comparison between more than two design alternatives
 - Ranking and selection of design alternatives
- Screening design
- Response surface design: Optimal design of the system
 - Stochastic approximation
 - Alternative Heuristic optimisation: Simulated annealing

Introduction

- Design of a system
 - Formally, a design α can be determined completely by one or more parameters $(v_1, v_2, \dots, v_k)$, e.g.
 - Maximum capacity of inventory
 - Economic order quantity
 - Number, speed or quality of machine, . . .
 - Replenishment policy, priority policy
 - . . .
 - Design space: set $\mathcal{A}$ of all feasible values α . Each design has a relevant cost $J(\alpha)$ relating to aspects of
 - Implementation
 - Desirability, acceptability, . . .
 - Performance => Evaluated by Discrete Event Simulation

Introduction

- Design of a system

- Trajectories of Markov Chains are dependent on the design of a system

- A family of Markov Chains $\chi(\alpha) = \{Y_k \in \Omega\}$ is a sequence of random variables, recursively defined by

$$Y_{k+1}(\alpha) = \phi(Y_k(\alpha), X_k, \alpha)$$

where $\{X_k \in \Theta\}$ is a sequence of independent and identically distributed random variables

$\phi: \Omega \times \Theta \times \mathcal{A} \rightarrow \Omega$ is a deterministic function

$Y_0(\alpha)$ is an independent random variable

Introduction

- The design problem

- Description: Select an appropriate value for $\alpha \in \mathcal{A}$

- Minimise cost (financial, ecological, ...)
 - Maximise gain or utility (e.g. Maximise throughput at traffic lights)
 - Minimum or maximum threshold values

- => Trade-off required based on conflicting objectives

- Formulation

$$\alpha^* = \arg \min_{\alpha \in \mathcal{A}} J(\alpha)$$

Introduction

- Estimation of the cost

➤ Cost $J(\alpha)$ must be estimated from a generated trajectory Y_k ($k = 1, \dots, K$)

$$J(\alpha) = E[h(\alpha, Y(\alpha))]$$

where α the design of the system

$h(\alpha, x)$ the cost function

$F_Y(\alpha, x)$ the distribution function of the random numbers $Y(\alpha)$

Introduction

- Estimation of the cost

- Integration

$$J(\alpha) = E[h(\alpha, Y(\alpha))] = \int_{x \in \Omega} h(\alpha, x) f(\alpha, x) dx$$

where $f(\alpha, x)$ the known probability distribution function of the random numbers $Y(\alpha)$

- Monte Carlo estimation

$$\bar{J}(\alpha) = \frac{1}{K} \sum_{k=1}^K h(\alpha, Y_k(\alpha))$$

- When the integral is too complex to calculate analytically or numerically
 - When the behaviour of the system depends on the time evolution of the trajectory
 - => Dynamic model: Use Markov Chains and Discrete Event Simulation to generate trajectories $Y_k(\alpha)$

Experimental design

- Experimental design terminology

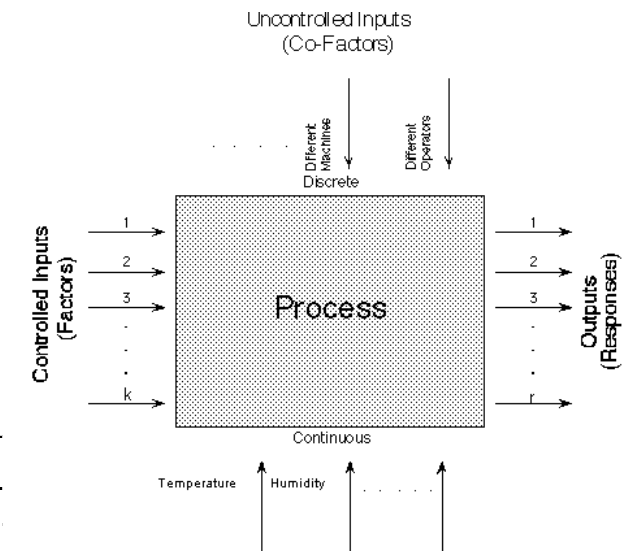
- Input parameters and structural assumptions composing a model are called factors

- Quantitative (or numerical) vs qualitative (or categorical) factors
- Controllable (focus) vs uncontrollable factors

- Output performance measures are called responses

- Example

System	Possible factors	Quantitative or qualitative?	Possible responses
Inventory system	Reorder point	Quantitative	Average cost per month, average number of items in inventory
	Order-up-to level	Quantitative	
Manufacturing system	Number of machines	Quantitative	Average time in system, throughput, average queue size
	Number of forklifts	Quantitative	
	Forklift dispatching rule	Qualitative	
Communications network	Number of nodes	Quantitative	Average end-to-end delay, throughput
	Number of links	Quantitative	
	Routing policy	Qualitative	



Experimental design

- The choice of an experimental design depends on
 - the objectives of the experiment
 - Comparative objective: The goal of the *comparative design* experiment is to make a conclusion about one a-priori important factor and to determine whether or not the factor is significant, i.e., whether or not there is a significant change in the response for different levels of that factor.
 - Screening objective: The purpose of the *main effects design* experiment is to select or *screen out* the few important main effects from the many less important ones.
 - Response Surface (method) objective: The *response surface method (RSM) design* experiment allows to estimate interaction and even quadratic effects (e.g. via regression analysis). This gives an idea of the (local) shape/mathematical function of the response surface, which can be optimised.
 - the number of factors to be investigated

Experimental design

- (notes on the blackboard – not in your slides)

- Screening design: Linear regression model

Linear model: $J(\alpha) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2$ (number of factors $k = 2$)

Linear model: $J(\alpha) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_{12} x_1 x_2 + \beta_{13} x_1 x_3 + \beta_{23} x_2 x_3 + \beta_{123} x_1 x_2 x_3$
(number of factors $k = 3$)

- Response surface design: Linear and quadratic regression model

Linear model: $J(\alpha) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2$ (number of factors $k = 2$)

Second order model with interaction (number of factors = 2)

$$J(\alpha) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{11} x_1^2 + \beta_{22} x_2^2 + \beta_{12} x_1 x_2$$

=> Relationship/dynamics between models: see next week!

Experimental design

- The choice of an experimental design

Number of factors	Comparative Objective	Screening objective	Response Surface Objective
1	1-factor completely randomised design	-	-
2-4	Randomised block design	Full or fractional factorial	Central Composite or Box-Behnken
5 or more	Randomised block design	Fractional factorial or Placket Burman	Screen first to reduce number of factors

Reference

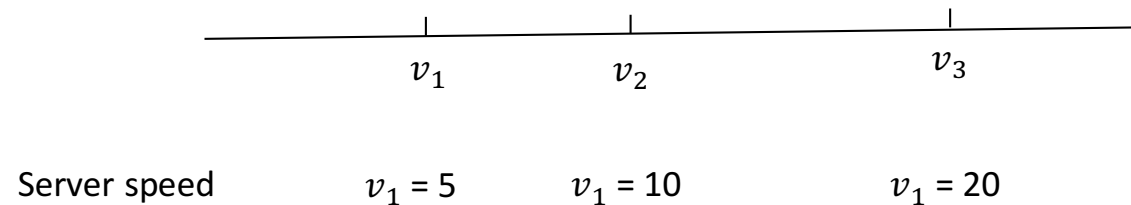
Montgomery, D.C., 2013, Design and Analysis of Experiments, 8th Edition, John Wiley, New York

Comparison between design alternatives

- Comparison Design Methodology
 - 1-factor completely randomised design
 - Number of factors = 1
 - The experiment compares the values of a response variable based on different levels of the primary factor
 - Number of independent simulation runs = Number of levels x Number of replications
 - Randomised block design
 - Number of factors = $k \geq 2$
 - The experiment compares the values of a response variable based on different designs, which result from the specific settings of the factors under consideration
 - Number of independent simulation runs (2 factor block design) = Number of levels factor 1 x Number of levels factor 2 x Number of replications
 - We assume that the number of designs is limited, otherwise screening is required first

Comparison between design alternatives

- (notes on the blackboard – not in your slides)

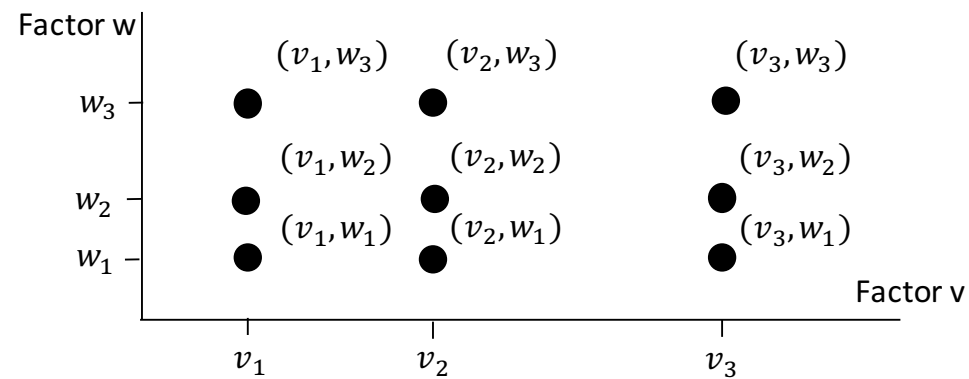


Comparison between design alternatives

- Comparison Design Methodology
 - 1-factor completely randomised design
 - Number of factors = 1
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Comparison between design alternatives

- (notes on the blackboard – not in your slides)



Comparison between design alternatives

- Decide between design alternatives α_1 and α_2 via Monte Carlo Estimation
 - Which of the designs $\alpha_1, \alpha_2 \in \mathcal{A}$ has the lowest cost?
 - Estimate the cost difference: $J(\alpha_1) - J(\alpha_2)$
 - Method
 - Use a simpler model (approximation) to calculate analytical results
 - When the resulting difference is too small, use Monte Carlo estimation
 - Independent simulation of both designs
 - Generate $Y_k(\alpha_1)$, $k = 1, \dots, K$ and calculate the Monte Carlo estimate: $\bar{J}(\alpha_1) = \frac{1}{K} \sum_{k=1}^K h(\alpha, Y_k(\alpha_1))$
 - Generate $Y_k(\alpha_2)$, $k = 1, \dots, K$ and calculate the Monte Carlo estimate: $\bar{J}(\alpha_2) = \frac{1}{K} \sum_{k=1}^K h(\alpha, Y_k(\alpha_2))$
- ⇒ Variance of the difference: $\text{var}[\bar{J}(\alpha_1) - \bar{J}(\alpha_2)] = \text{var}[\bar{J}(\alpha_1)] + \text{var}[\bar{J}(\alpha_2)] - 2\text{cov}[\bar{J}(\alpha_1), \bar{J}(\alpha_2)]$
- ⇒ Relative error of the estimate: $\frac{\sqrt{\text{var}[\bar{J}(\alpha_1)] + \text{var}[\bar{J}(\alpha_2)] - 2\text{cov}[\bar{J}(\alpha_1), \bar{J}(\alpha_2)]}}{\bar{J}(\alpha_1) - \bar{J}(\alpha_2)}$ which is very large due to small difference $\bar{J}(\alpha_1) - \bar{J}(\alpha_2)$ and $\text{cov}[\bar{J}(\alpha_1), \bar{J}(\alpha_2)] = 0$ (experiments are independent)

Comparison between design alternatives

- Decide between design alternatives α_1 and α_2 via Monte Carlo Estimation

- Independent simulation of both designs

- Sometimes the only way to compare different designs (especially if number of observations design 1 (n_1) $\neq$ number of observations design 2 (n_2))

For example: Assume that system 1 is the real-world system on which we have physically collected data and system 2 is a simulation model, it is likely that $n_1 < n_2$

- Use the confidence interval of Welch (1938)

$$\bar{X}_i(n_i) = \frac{1}{n_i} \sum_{j=1}^{n_i} X_{ij}$$

$$S_i^2(n_i) = \frac{1}{n_i - 1} \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_i)^2$$

$$\hat{f} = \frac{[S_1^2(n_1)/n_1 + S_2^2(n_2)/n_2]^2}{[S_1^2(n_1)/n_1]^2/(n_1 - 1) + [S_2^2(n_2)/n_2]^2/(n_2 - 1)}$$

$$1 - \alpha \text{ confidence interval: } [\bar{X}_1(n_1) - \bar{X}_2(n_2) - t_{\hat{f}, 1 - \alpha/2} \sqrt{S_1^2(n_1)/n_1 + S_2^2(n_2)/n_2},$$

$$\bar{X}_1(n_1) - \bar{X}_2(n_2) + t_{\hat{f}, 1 - \frac{\alpha}{2}} \sqrt{S_1^2(n_1)/n_1 + S_2^2(n_2)/n_2}]$$

Comparison between design alternatives

- (on the blackboard – not in your slides)
- Decide between design alternatives α_1 and α_2 via Monte Carlo Estimation
 - Which of the designs $\alpha_1, \alpha_2 \in \mathcal{A}$ has the lowest cost?

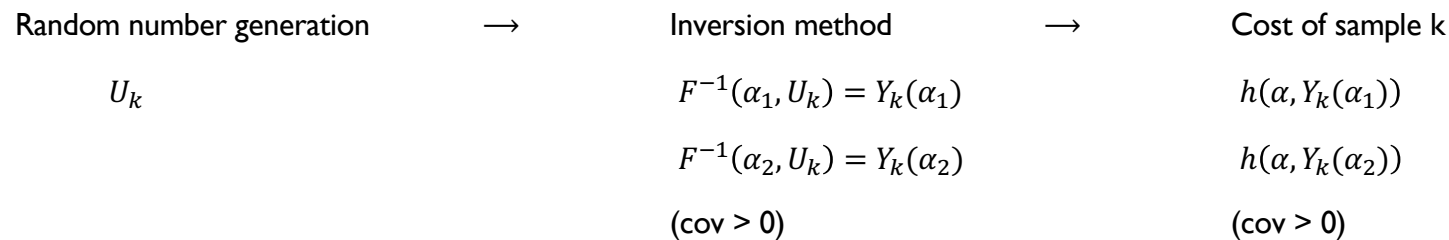
⇒ Relative error of the estimate: $\frac{\sqrt{\text{var}[\bar{J}(\alpha_1)] + \text{var}[\bar{J}(\alpha_2)] - 2\text{cov}[\bar{J}(\alpha_1), \bar{J}(\alpha_2)]}}{\bar{J}(\alpha_1) - \bar{J}(\alpha_2)}$

Comparison between design alternatives

- Decide between design alternatives α_1 and α_2 via Monte Carlo Estimation

➤ Dependent simulation of both designs

- Assumption: the number of observations is the same for both designs (or we deleted observations for one design)
- Generate trajectories with strong positive correlation ($\text{cov}[\bar{J}(\alpha_1), \bar{J}(\alpha_2)] \gg 0$) between the two estimators $\bar{J}(\alpha_1)$ and $\bar{J}(\alpha_2)$ such that relative error decreases
- Method: Common Random Number Generation
 - Use the same random numbers for both trajectories
 - Principle



(Reduced variance due to positive covariance)

Comparison between design alternatives

- Decide between design alternatives α_1 and α_2 via Monte Carlo Estimation

➤ Dependent simulation of both designs

- Paired-t confidence interval: $[\bar{\bar{\Delta}} - \frac{t_{n-1,1-\alpha/2}^S}{\sqrt{n}}, \bar{\bar{\Delta}} + \frac{t_{n-1,1-\alpha/2}^S}{\sqrt{n}}]$ with $\bar{\bar{\Delta}} = \bar{J}(\alpha_1) - \bar{J}(\alpha_2)$
- Student t-distribution

TABLE of CRITICAL VALUES for STUDENT'S t DISTRIBUTIONS												
Column headings denote probabilities (α) above tabulated values.												
d.f.	0.40	0.25	0.10	0.05	0.04	0.025	0.02	0.01	0.005	0.0025	0.001	0.0005
1	0.325	1.000	3.078	6.314	7.916	12.706	15.894	31.821	63.656	127.321	318.289	636.578
2	0.289	0.816	1.886	2.920	3.320	4.303	4.849	6.965	9.925	14.089	22.328	31.600
3	0.277	0.765	1.638	2.353	2.605	3.182	3.482	4.541	5.841	7.453	10.214	12.924
4	0.271	0.741	1.533	2.132	2.333	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	0.267	0.727	1.476	2.015	2.191	2.571	2.757	3.365	4.032	4.773	5.894	6.869
6	0.265	0.718	1.440	1.943	2.104	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	0.263	0.711	1.415	1.895	2.046	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	0.262	0.706	1.397	1.860	2.004	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	0.261	0.703	1.383	1.833	1.973	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	0.260	0.700	1.372	1.812	1.948	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	0.260	0.697	1.363	1.796	1.928	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	0.259	0.695	1.356	1.782	1.912	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	0.259	0.694	1.350	1.771	1.899	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	0.258	0.692	1.345	1.761	1.887	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	0.258	0.691	1.341	1.753	1.878	2.131	2.249	2.602	2.947	3.286	3.733	4.073
16	0.258	0.690	1.337	1.746	1.869	2.120	2.235	2.583	2.921	3.252	3.686	4.015
17	0.257	0.689	1.333	1.740	1.862	2.110	2.224	2.567	2.898	3.222	3.646	3.965
18	0.257	0.688	1.330	1.734	1.855	2.101	2.214	2.552	2.878	3.197	3.610	3.922
19	0.257	0.688	1.328	1.729	1.850	2.093	2.205	2.539	2.861	3.174	3.579	3.883
20	0.257	0.687	1.325	1.725	1.844	2.086	2.197	2.528	2.845	3.153	3.552	3.850
21	0.257	0.686	1.323	1.721	1.840	2.080	2.189	2.518	2.831	3.135	3.527	3.819
22	0.256	0.686	1.321	1.717	1.835	2.074	2.183	2.508	2.819	3.119	3.505	3.792
23	0.256	0.685	1.319	1.714	1.832	2.069	2.177	2.500	2.807	3.104	3.485	3.768
24	0.256	0.685	1.318	1.711	1.828	2.064	2.172	2.492	2.797	3.091	3.467	3.745
25	0.256	0.684	1.316	1.708	1.825	2.060	2.167	2.485	2.787	3.078	3.450	3.725
26	0.256	0.684	1.315	1.706	1.822	2.056	2.162	2.479	2.779	3.067	3.435	3.707
27	0.256	0.684	1.314	1.703	1.819	2.052	2.158	2.473	2.771	3.057	3.421	3.689
28	0.256	0.683	1.313	1.701	1.817	2.048	2.154	2.467	2.763	3.047	3.408	3.674
29	0.256	0.683	1.311	1.699	1.814	2.045	2.150	2.462	2.756	3.038	3.396	3.660
30	0.256	0.683	1.310	1.697	1.812	2.042	2.147	2.457	2.750	3.030	3.385	3.646
31	0.256	0.682	1.309	1.696	1.810	2.040	2.144	2.453	2.744	3.022	3.375	3.633
32	0.255	0.682	1.309	1.694	1.808	2.037	2.141	2.449	2.738	3.015	3.365	3.622
33	0.255	0.682	1.308	1.692	1.806	2.035	2.138	2.445	2.733	3.008	3.356	3.611
34	0.255	0.682	1.307	1.691	1.805	2.032	2.136	2.441	2.728	3.002	3.348	3.601
35	0.255	0.682	1.306	1.690	1.803	2.030	2.133	2.438	2.724	2.996	3.340	3.591
36	0.255	0.681	1.306	1.688	1.802	2.028	2.131	2.434	2.719	2.990	3.333	3.582
37	0.255	0.681	1.305	1.687	1.800	2.026	2.129	2.431	2.715	2.985	3.326	3.574
38	0.255	0.681	1.304	1.686	1.799	2.024	2.127	2.429	2.712	2.980	3.319	3.566
39	0.255	0.681	1.304	1.685	1.798	2.023	2.125	2.426	2.708	2.976	3.313	3.558
40	0.255	0.681	1.303	1.684	1.796	2.021	2.123	2.423	2.704	2.971	3.307	3.551
60	0.254	0.679	1.296	1.671	1.781	2.000	2.099	2.390	2.660	2.915	3.232	3.460
80	0.254	0.678	1.292	1.664	1.773	1.990	2.088	2.374	2.639	2.887	3.195	3.416
100	0.254	0.677	1.290	1.660	1.769	1.984	2.081	2.364	2.626	2.871	3.174	3.390
120	0.254	0.677	1.289	1.658	1.766	1.980	2.076	2.358	2.617	2.860	3.160	3.373
140	0.254	0.676	1.288	1.656	1.763	1.977	2.073	2.353	2.611	2.852	3.149	3.361
160	0.254	0.676	1.287	1.654	1.762	1.975	2.071	2.350	2.607	2.847	3.142	3.352
180	0.254	0.676	1.286	1.653	1.761	1.973	2.069	2.347	2.603	2.842	3.136	3.345
200	0.254	0.676	1.286	1.653	1.760	1.972	2.067	2.345	2.601	2.838	3.131	3.340
250	0.254	0.675	1.285	1.651	1.758	1.969	2.065	2.341	2.596	2.832	3.123	3.330
inf	0.253	0.674	1.282	1.645	1.751	1.960	2.054	2.326	2.576	2.807	3.090	3.290

Comparison between design alternatives

- Decide between design alternatives α_1 and α_2 via Monte Carlo Estimation

➤ Example 1: The Automated Bank Teller Problem

- Assume a bank is about to install and operate an automated bank teller and the bank has to choose out of the following alternative designs

- Alternative 1: One Zippytel machine - This machine costs twice as much but works twice as fast as one Klunkytel machine
- Alternative 2: Two Klunkytel machines

=> The total cost for the bank is the same, the bank would like to implement the system that provides the best service

- Input parameters

- Arrival pattern of customers $\sim$ Poisson distribution (λ) with $\lambda = 1$ (i.e. one arrival per minute)
- Service pattern of bank tellers

Alternative 1: Zippytel machine - Exponential distribution (μ) with $1/\mu = 0.9$ (mean service time) – Utilisation $\rho = \frac{\lambda}{\mu} = \frac{1}{10/9} = 0.9$

Alternative 2: Klunkytel machine - Exponential distribution (μ) with $1/\mu = 1.8$ (mean service time) – Utilisation $\rho = \frac{\lambda}{s\mu} = \frac{1}{2 \times 10/18} = 0.9$

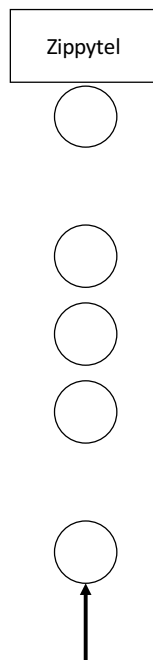
Comparison between design alternatives

- Decide between design alternatives α_1 and α_2 via Monte Carlo Estimation

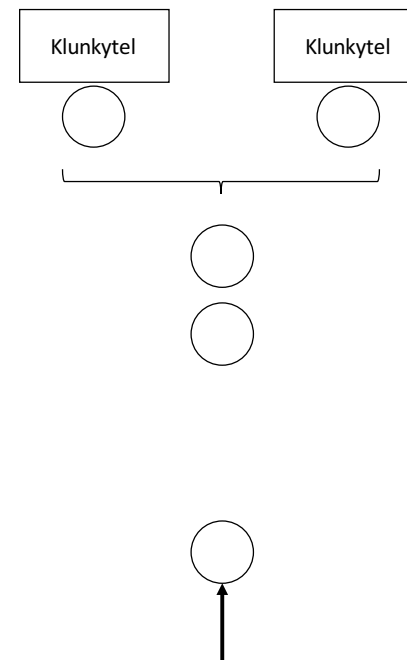
➤ Example I: The Automated Bank Teller Problem

- System configuration

Alternative 1: M/M/1



Alternative 2: M/M/2



Comparison between design alternatives

- Decide between design alternatives α_1 and α_2 via Monte Carlo Estimation

➤ Example I: The Automated Bank Teller Problem

- Performance measure: Expected average queueing delay for the first 100 customers

$W_{q,Z}(100)$ Waiting time alternative 1 (one Zippytel machine)

$W_{q,K}(100)$ Waiting time alternative 2 (two Klunkytel machines)

=> Compare waiting times to infer whether $W_{q,Z}(100)$ or $W_{q,K}(100)$ is smaller starting from 0 customers

- Analytical results: Steady state-behaviour

$$W_{q,Z}(100) = \frac{\lambda}{\mu(\mu-\lambda)} = \frac{1}{10/9(10/9-1)} = 8.1 \quad (\text{Queueing results – M/M/1 Queue}) \quad (\text{Gross et al., 2008})$$

$$W_{q,K}(100) = \frac{(\lambda/\mu)^2 \mu}{(k-1)!(k\mu-\lambda)^2} P_0 = 7.67 \quad (\text{Queueing results – M/M/2 Queue with } P_0 = \frac{1}{\sum_{n=0}^{k-1} \frac{1}{n!} (\lambda/\mu)^n + \frac{1}{k!} (\lambda/\mu)^k (k\mu/\mu - \lambda)})$$

- Analytical results: First 100 customers (Transient State)

$$W_{q,Z}(100) = 4.13 \quad W_{q,K}(100) = 3.70 \quad (\text{Kelton and Law, 1985})$$

References

Gross, D., Shortle, J.F., Thompson, J.M. And Harris, C.M., 2008, Fundamentals of Queueing Theory (4th ed.), 528pp., Wiley, New Jersey (US).

Kelton, W.D. And Law, A.M., 1985, The transient behaviour of the M/M/s Queue with Implications for Steady-State Simulation, Operations Research, 33, 378-396.

Comparison between design alternatives

- Decide between design alternatives α_1 and α_2 via Monte Carlo Estimation

➤ Example I: The Automated Bank Teller Problem

- Independent simulation of both designs: 100 independent pairs of simulation runs

$\bar{W}_{q,Z,i}(100)$ Waiting time of first 100 customers for simulation run i (Alternative 1)

$\bar{W}_{q,K,i}(100)$ Waiting time of first 100 customers for simulation run i (Alternative 2)

- Results: Pairwise run comparison

Experiment	$\bar{W}_{q,Z,n}(100)$	$\bar{W}_{q,K,n}(100)$	Recommendation
1	8.33	3.60	M/M/2 - Klunkytel (right)
2	1.24	5.66	M/M/1 - Zippytel (wrong)
3	1.73	2.84	M/M/1 - Zippytel (wrong)
4	5.99	6.97	M/M/1 - Zippytel (wrong)
5	2.29	2.35	M/M/1 - Zippytel (wrong)
6	3.36	13.19	M/M/1 - Zippytel (wrong)
7	3.87	1.74	M/M/2 - Klunkytel (right)
...			
98	1.12	5.27	M/M/1 - Zippytel (wrong)
99	2.10	5.35	M/M/1 - Zippytel (wrong)
100	0.70	6.24	M/M/1 - Zippytel (wrong)

In 45 out of 100 experiments
 $\bar{W}_{q,K,n}(100) < \bar{W}_{q,Z,n}(100)$

=> Difficult to make the
 right conclusions

Comparison between design alternatives

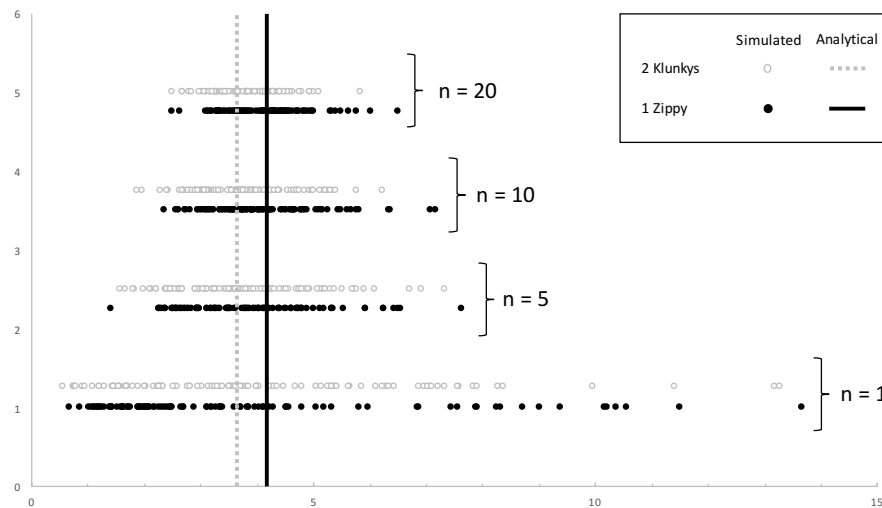
- Decide between design alternatives α_1 and α_2 via Monte Carlo Estimation

➤ Example I: The Automated Bank Teller Problem

- Independent simulation of both designs: 100 independent pairs of simulation runs
 - Assume each run is composed out of n replications and calculate the average, i.e.

$$\bar{W}_{q,Z,i}(100) = \frac{1}{n} \sum_{j=1}^n \tilde{W}_{q,Z,i,n}(100) \text{ and } \bar{W}_{q,K,i}(100) = \frac{1}{n} \sum_{j=1}^n \tilde{W}_{q,K,i,n}(100)$$

- Results: Average delay in queue



Proportion of experiments	
n	favoring the one-Zippy system
1	0.55
5	0.48
10	0.40
20	0.40

Probability for making errors decreases when n increases but there is still a significant error

Comparison between design alternatives

- Decide between design alternatives α_1 and α_2 via Monte Carlo Estimation

➤ Example I: The Automated Bank Teller Problem

- Independent simulation of both designs
- Evaluate the cost difference: $\bar{\Delta}_i = \bar{W}_{q,Z,i}(100) - \bar{W}_{q,K,i}(100)$ for each simulation run i
- Compute the mean difference: $\bar{\bar{\Delta}} = \frac{1}{100} \sum_{i=1}^{100} \bar{\Delta}_i = -0.37$

$$\text{Welch confidence interval: } \bar{\bar{\Delta}} \in [\bar{X}_1(n_1) - \bar{X}_2(n_2) - t_{\hat{f}, 1-\alpha/2} \sqrt{S_1^2(n_1)/n_1 + S_2^2(n_2)/n_2}, \\ \bar{X}_1(n_1) - \bar{X}_2(n_2) + t_{\hat{f}, 1-\alpha/2} \sqrt{S_1^2(n_1)/n_1 + S_2^2(n_2)/n_2}]$$

$$\Rightarrow \bar{\bar{\Delta}} \in [-1.12; 0.39]$$

$$\text{with } S_1 = 2.80 \text{ and } S_2 = 2.63$$

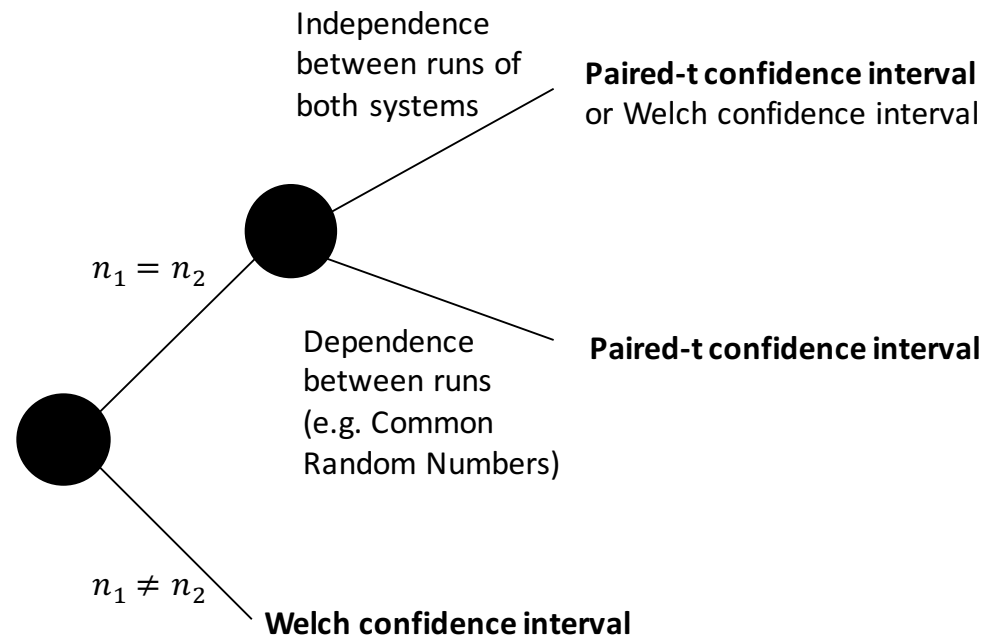
$$t_{\hat{f}, 1-\alpha/2} = 1.97 \text{ (95\%-confidence interval) and } \hat{f} = 197.25$$

$$n = 100$$

=> There is no statistical difference between both systems as 0 lies within this interval

Comparison between design alternatives

- (note on the blackboard – not in your slides)



Comparison between design alternatives

- Decide between design alternatives α_1 and α_2 via Monte Carlo Estimation

➤ Example I: The Automated Bank Teller Problem

- Independent simulation of both designs
- Evaluate the cost difference: $\bar{\Delta}_i = \bar{W}_{q,Z,i}(100) - \bar{W}_{q,K,i}(100)$ for each simulation run i
- Compute the mean difference: $\bar{\bar{\Delta}} = \frac{1}{100} \sum_{i=1}^{100} \bar{\Delta}_i = -0.37$

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$$\Rightarrow \bar{\bar{\Delta}} \in [-1.12; 0.39]$$

$$\text{with } S_1 = 2.80 \text{ and } S_2 = 2.63$$

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$$n = 100$$

=> There is no statistical difference between both systems as 0 lies within this interval

Comparison between design alternatives

- Decide between design alternatives α_1 and α_2 via Monte Carlo Estimation

➤ Example I: The Automated Bank Teller Problem

- Dependent simulation of both designs via Common Random Numbers

- Principle

	Random number generation	→	Inversion method	→	Performance measure
Arrival time customer k	U_k		$F^{-1}(\alpha_1, U_k) = A_k(\alpha_1)$ $F^{-1}(\alpha_2, U_k) = A_k(\alpha_2)$		
Service time customer k	U'_k		$F^{-1}(\alpha_1, U'_k) = S_k(\alpha_1)$ $F^{-1}(\alpha_2, U'_k) = S_k(\alpha_2)$		
Waiting time					$Y_{k+1} = \max(Y_k + S_k - A_k, 0)$

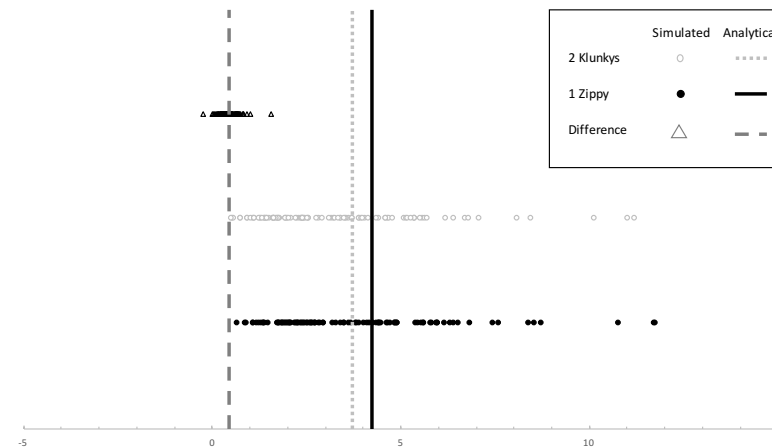
Comparison between design alternatives

- Decide between design alternatives α_1 and α_2 via Monte Carlo Estimation

➤ Example I: The Automated Bank Teller Problem

- Dependent simulation of both designs via Common Random Numbers
 - Evaluate the cost difference: $\bar{\Delta}_i = \bar{W}_{q,Z,i}(100) - \bar{W}_{q,K,i}(100)$ for each simulation run i

Experiment	$W_{q,Z,n}(100)$	$W_{q,K,n}(100)$	Δ_i
1	3.55	2.88	0.67
2	1.24	5.66	0.43
3	1.73	2.84	0.23
4	5.99	6.97	0.40
5	2.29	2.35	0.61
6	3.36	13.19	0.41
7	3.87	1.74	0.42
...			
98	1.12	5.27	0.08
99	2.10	5.35	0.41
100	0.70	6.24	0.40
	3.81	3.37	0.45



- Compute the mean difference: $\bar{\Delta} = \frac{1}{100} \sum_{i=1}^{100} \bar{\Delta}_i = 3344.87$
 - 95% confidence interval: $\bar{\Delta} \in \left[\bar{\Delta} - \frac{t_{n-1,1-\alpha/2} S}{\sqrt{n}}, \bar{\Delta} + \frac{t_{n-1,1-\alpha/2} S}{\sqrt{n}} \right] = [0.398; 0.493]$ => Statistical difference between both systems
- with $S = 0.238$ and $t_{n-1,1-\alpha/2} = 1.98421$ (95%-confidence interval: $n = \text{\#runs (100)}$; $\alpha = 0.05$)

Comparison between design alternatives

- Decide between design alternatives α_1 and α_2 via Monte Carlo Estimation

➤ Example 2: The Jobshop problem: Comparing scheduling strategies

- Assume a flexible production problem with
 - P product types: Arrival pattern $\sim$ Poisson (λ_p)
 - M different machines: Service time of kth item of type p $\sim F_S(s|k,m,p)$
 - Scheduling strategy STR_i : Set of rules for allocating tasks to machines (e.g. minimise maximal queue length, balance queue lengths, ...)
- Which of the two scheduling strategies STR_1 and STR_2 results in the lowest cost?
 - Compare $J(STR_1)$ and $J(STR_2)$ by running two simulations and estimate $\bar{J}(STR_1)$ and $\bar{J}(STR_2)$
 - Generate two trajectories for STR_1 and STR_2
 - => Use the same arrival times
 - => Ensure a positive correlation of service time (if item k of type p is executed on machine m under STR_1 and on machine m' under STR_2)

Random number generation

→

Inversion method

$$U_{k,p}$$

$$S_{k,m,p} = F_S^{-1}(U_{k,p}|k, m, p) \quad \text{for } STR_1$$

$$S_{k,m',p} = F_S^{-1}(U_{k,p}|k, m', p) \quad \text{for } STR_2$$

$\bar{J}(STR_1)$ and $\bar{J}(STR_2)$ will be highly correlated estimates and their difference will have reduced variance

Comparison between design alternatives

- Decide between more than two design alternatives

➤ Principle

- Suppose that we make c confidence-intervals simultaneously. The probability that all c confidence intervals contain their true measure satisfies

$$P(\mu_s \in I_s \text{ for all } s = 1, 2, \dots, c) \geq 1 - \sum_{s=1}^c \alpha_s \quad (\text{Bonferroni inequality})$$

with μ_s true measure of performance for design s

I_s confidence interval for design s

α_s confidence level for design s

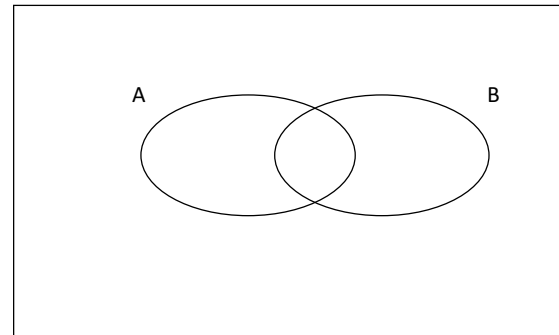
- The individual confidence intervals have to be adjusted upward so that the overall confidence level of all intervals is at least the desired level $1 - \alpha$, i.e. each separate confidence interval should consider the confidence level $1 - \alpha/c$
 - For example: If $c = 10$ and $\alpha = 0.90$, we must make each individual interval at the 99 percent level.
 - For large c , the individual intervals may become quite wide

Comparison between design alternatives

- (note on the blackboard – not in your slides)

➤ Bonferroni inequality

- Assume A and B are two events testing the null hypothesis and assume that both nulls are true
 - A: null hypothesis 1 is rejected
 - B: null hypothesis 2 is rejected



- $\Pr[A \cup B] = \Pr[A] + \Pr[B] - \Pr[A \cap B]$
- $\Pr[A \cup B] \leq \Pr[A] + \Pr[B]$
- $1 - \Pr[A \cup B] \geq 1 - \Pr[A] - \Pr[B]$

Assume $\Pr[A] = \alpha_1$ (probability that confidence interval 1 is rejected)

$\Pr[B] = \alpha_2$ (probability that confidence interval 2 is rejected)

$\Pr[A \cup B] = \alpha$ (probability that confidence interval 1 and/or 2 is rejected)

or $1 - \alpha \geq 1 - \alpha_1 - \alpha_2$

or overall confidence level ≥ 1 minus sum of probabilities that individual hypotheses are rejected

Comparison between design alternatives

- Decide between more than two design alternatives

➤ Principle

- Suppose that we make c confidence-intervals simultaneously. The probability that all c confidence intervals contain their true measure satisfies

$$P(\mu_s \in I_s \text{ for all } s = 1, 2, \dots, c) \geq 1 - \sum_{s=1}^c \alpha_s \quad (\text{Bonferroni inequality})$$

with μ_s true measure of performance for design s

I_s confidence interval for design s

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- The individual confidence intervals have to be adjusted upward so that the overall confidence level of all intervals is at least the desired level $1 - \alpha$, i.e. each separate confidence interval should consider the confidence level $1 - \alpha/c$
 - For example: If $c = 10$ and $\alpha = 0.90$, we must make each individual interval at the 99 percent level.
 - For large c , the individual intervals may become quite wide

Comparison between design alternatives

- Decide between more than two design alternatives

- Comparison with a standard design

- Suppose that one design is a 'standard' (e.g. the existing system or policy)
- Construct $k-1$ confidence intervals that compares an alternative system with the standard design,

i.e. $\mu_2 - \mu_1, \mu_3 - \mu_1, \dots, \mu_k - \mu_1$

with μ_1 The true value of the performance measure for the standard design

μ_s The true value of the performance measure for an alternative design ($s = 2, \dots, k$)

- In order to establish a confidence level of at least $1 - \alpha$, each individual interval should be constructed at the level $1 - \alpha/(k-1)$ ($c = k-1$)

=> Then we can say with a confidence level $1 - \alpha$ that for all $s = 2, 3, \dots, k$,

design s differs from the standard design if the interval for $\mu_s - \mu_1$ misses 0

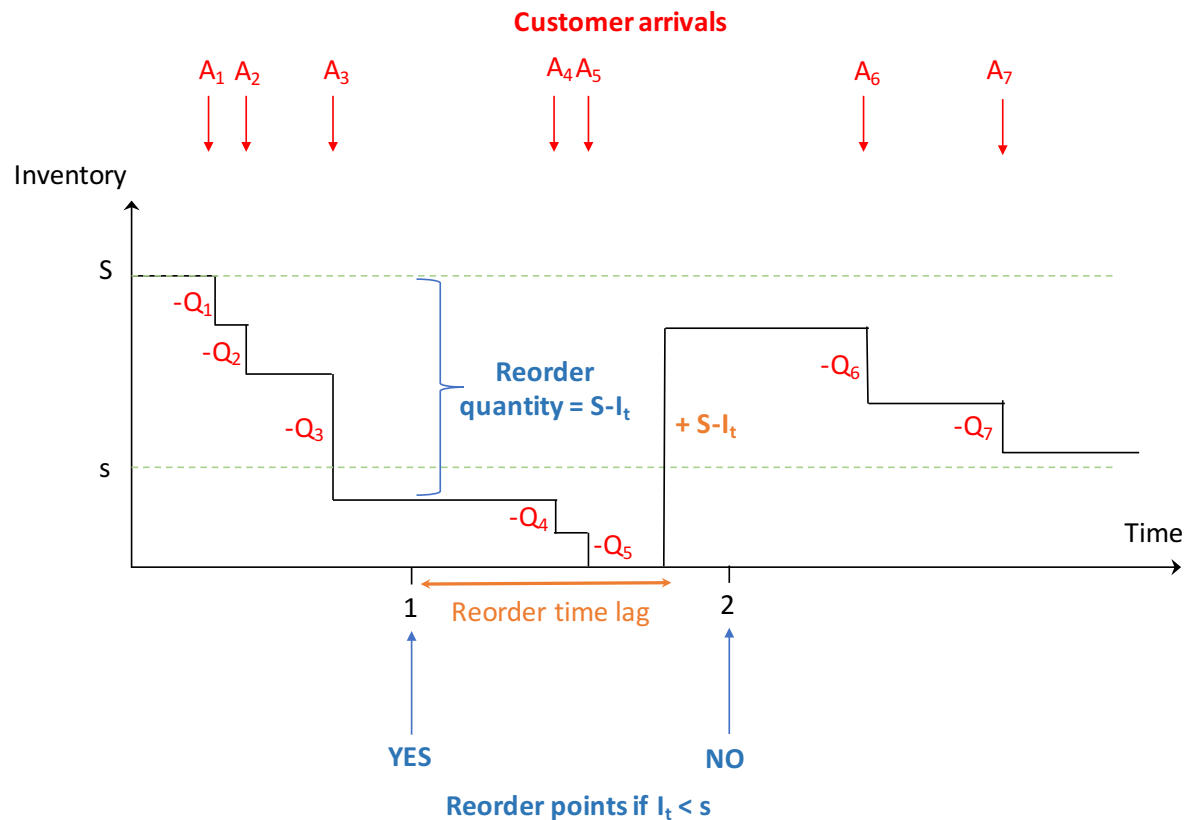
design s is not significantly different from the standard if this interval contains 0

Comparison between design alternatives

- Decide between more than two design alternatives
 - Comparison with a standard design – Example: An Inventory Problem
 - A shop sells a certain product to customers (available in discrete unit measurements)
 - Customers arrive at the shop according to a Poisson process with parameter λ and purchase a certain quantity of the product according to a discrete random variable with an arbitrary probability mass function
 - If the shop has less units available than the customer demand, the customer buys all remaining items but no backordering occurs
 - The selling price of the product equals \$d per unit
 - Inventory control and reordering policy
 - The number of units in inventory in the shop available for sale is the shop inventory
 - Holding cost is a linear function of time
 - When inventory falls below a certain threshold, an order is placed to a supplier
 - (s, S) ordering policy: When the number of units available falls to s or less, a sufficient quantity is ordered to bring the amount in stock up to S

Comparison between design alternatives

- (on the blackboard – not in your slides)
 - Comparison with a standard design – Example: An Inventory Problem



Comparison between design alternatives

- Decide between more than two design alternatives
 - Comparison with a standard design – Example: An Inventory Problem
 - Product inventory
 - For each item purchased from the supplier a variable cost of \$c per unit is incurred
 - Per order a fixed cost is incurred to cover transportation costs
 - The ordered items arrive with a delay between sending the order and received the units according to a general probability distribution $G(t)$
 - Only one order is given at a time and the shopkeeper waits until that order has been filled before sending an additional order
 - Objective
 - Determine which policies differ from the standard ordering policy $(s, S) = (20, 40)$

Policy (i)	s	S
1	20	40
2	20	80
3	40	60
4	40	100
5	60	100

Comparison between design alternatives

- Decide between more than two design alternatives
 - Comparison with a standard design – Example: An Inventory Problem
 - Results (n = 5 simulation runs)

j	X_{1j}	X_{2j}	X_{3j}	X_{4j}	X_{5j}
1	126.97	118.21	120.77	131.64	141.00
2	124.31	120.22	129.32	137.07	143.86
3	126.68	122.45	120.61	129.91	144.30
4	122.66	122.68	123.65	129.97	141.72
5	127.23	119.40	127.34	131.08	142.61
Mean	125.57	120.59	124.34	131.93	142.70
Variance	4.00	3.76	15.23	8.79	1.94

- 97.5%-Confidence intervals for all comparisons with the standard system

i	$\bar{X}_i - \bar{X}_1$	Paired-t		Welch	
		Half-length	Interval	Half-length	Interval
2	-4.98	5.46	[-10.44;0.48]	3.54	[-8.52;-1.44]*
3	-1.23	7.57	[-8.80;6.34]	6.20	[-7.44;4.97]
4	6.36	6.10	[0.27;12.46]*	4.54	[1.82;10.91]*
5	17.13	3.71	[13.42;20.83]*	3.10	[14.03;20.23]*

Use Common Random Numbers or increase number of replications to reduce CI

Comparison between design alternatives

- Decide between more than two design alternatives
 - All Pairwise Comparison
 - Compare each system with every other system to detect and quantify significant pairwise differences
 - Form confidence intervals for the differences $\mu_{i_2} - \mu_{i_1}$ for all i_1 and i_2 with $i_1 < i_2$
 - Construct $k(k-1)/2$ confidence intervals to compare each pair of alternative systems
 - => In order to establish a confidence level of at least $1 - \alpha$, each individual interval should be constructed at the level $1 - \alpha / (k(k-1)/2)$ ($c = k(k-1)/2$)

Comparison between design alternatives

- Decide between more than two design alternatives
 - All Pairwise Comparison – Example: An Inventory Problem
 - Since there are 10 possible pairs ($= 5/(5-1)/2$), each individual interval must be made with a 99% confidence interval in order to achieve 90 percent overall confidence
 - 99%-Confidence intervals for all pairwise comparisons (Paired-t confidence intervals)

$i_1 \backslash i_2$	2	3	4	5
1	-4.98±7.19	-1.23±9.97	6.36±8.03	17.13±4.88*
2		3.75±9.58	11.34±8.38*	22.11±3.78*
3			7.60±5.65*	18.36±7.71*
4				10.76±5.87*

Comparison between design alternatives

- Ranking and selection of design alternatives

- Principle

- Select one of the k systems as being the best one and to control the probability that the selected system is really the best one

- Assume

X_{ij} the performance measure value for the j^{th} replication of the i^{th} system (replications are independent)

μ_i the expected value of the performance measure, i.e. $\mu_i = E(X_{ij})$

μ_{i_l} the l^{th} smallest of the μ_i 's, so that $\mu_{i_1} \leq \mu_{i_2} \leq \dots \leq \mu_{i_k}$

- Goal: Select a system with the smallest expected response, i.e. μ_{i_1}
 - This event is denoted as the 'correct selection' (CS)
 - The randomness of the observed X_{ij} 's implies that we can never be sure to make the correct selection but we would like to prespecify the probability of CS
 - If μ_{i_1} and μ_{i_2} are very close, we might even not care if we erroneously select system i_2

Comparison between design alternatives

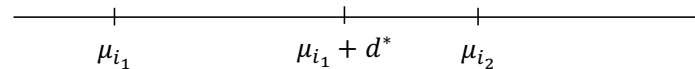
- Ranking and selection of design alternatives

- Problem formulation

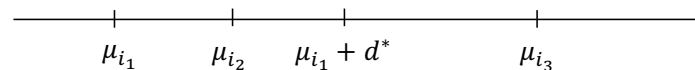
- $P(CS) \geq P^*$ provided that $\mu_{i_2} - \mu_{i_1} \geq d^*$

where $P^* > 1/k$ and the indifference amount d^* are both specified by the analyst

Choose system i_1 with probability P^*



Choose system i_1 or i_2 with probability P^*



=> The expected response of the selected system will be no larger than $\mu_{i_1} + d^*$ with probability P^*

Comparison between design alternatives

- Ranking and selection of design alternatives

- Solution method: Two-stage sampling procedure of Dudewicz and Dalal (1975)

- First stage

- Simulate a fixed number of replications ($n_0 \geq 2$) of each of the k design alternatives. n_0 should not be too small (not representative) or too large (time consuming), $n_0 = 20$ is a good estimate.
- Define the first stage sample means and variances

$$\bar{X}_i^{(1)}(n_0) = \frac{1}{n_0} \sum_{j=1}^{n_0} X_{ij} \quad \text{and} \quad S_i^2(n_0) = \frac{1}{n_0-1} \sum_{j=1}^{n_0} (X_{ij} - \bar{X}_i^{(1)}(n_0))^2 \quad \text{for } i = 1, 2, \dots, k$$

- Compute the total sample size N_i needed for system i to reach a decision as

$$N_i = \max\{n_0 + 1, \left\lceil \frac{h_1^2 S_i^2(n_0)}{(d^*)^2} \right\rceil\}$$

h_1 is a constant that depends on k, n_0 and P^* and can be obtained from the following table

P^*	n_0	k=2	k=3	k=4	k=5	k=6	k=7	k=8	k=9	k=10
0.90	20	1.896	2.342	2.583	2.747	2.870	2.969	3.051	3.121	3.182
0.90	40	1.852	2.283	2.514	2.669	2.785	2.878	2.954	3.019	3.076
0.95	20	2.453	2.872	3.101	3.258	3.377	3.472	3.551	3.619	3.679
0.95	40	2.386	2.786	3.003	3.150	3.260	3.349	3.422	3.484	3.539

Comparison between design alternatives

- Ranking and selection of design alternatives

- Solution method: Two-stage sampling procedure of Dudewicz and Dalal (1975)

- Second stage

- Simulate $N_i - n_0$ more of system i ($i = 1, 2, \dots, k$)
- Define the second stage sample means

$$\bar{X}_i^{(2)}(N_i - n_0) = \frac{1}{N_i - n_0} \sum_{j=n_0+1}^{N_i} X_{ij} \quad \text{for } i = 1, 2, \dots, k$$

- Compute the weights

$$W_{i1} = \frac{n_0}{N_i} \left\{ 1 + \sqrt{1 - \frac{N_i}{n_0} \left[1 - \frac{(N_i - n_0)(d^*)^2}{h_1^2 S_i^2(n_0)} \right]} \right\} \quad \text{and} \quad W_{i2} = 1 - W_{i1}$$

- Define the weighted sample means

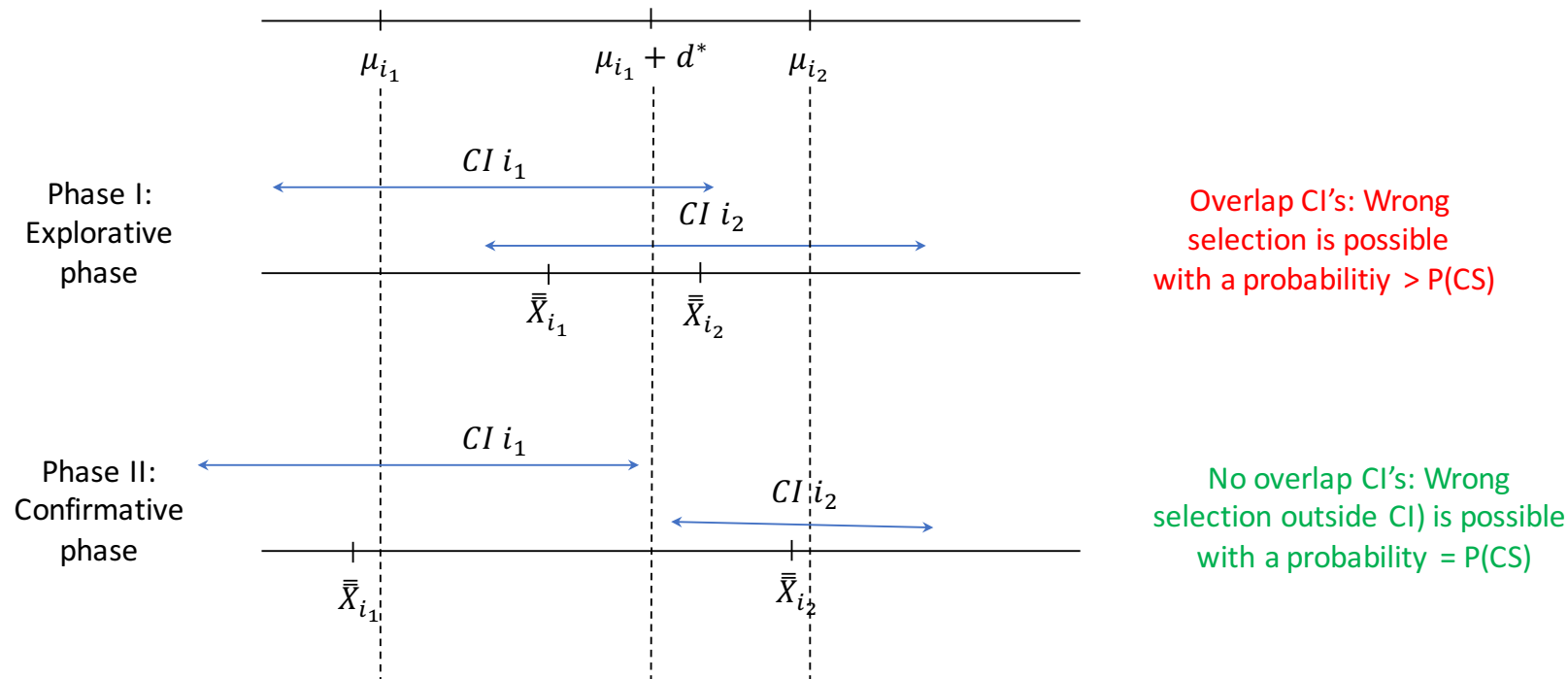
$$\bar{X}_i(N_i) = W_{i1} \bar{X}_i^{(1)}(n_0) + W_{i2} \bar{X}_i^{(2)}(N_i - n_0)$$

- Select the system with the smallest $\bar{X}_i(N_i)$

Comparison between design alternatives

- (on the blackboard – not in your slides)

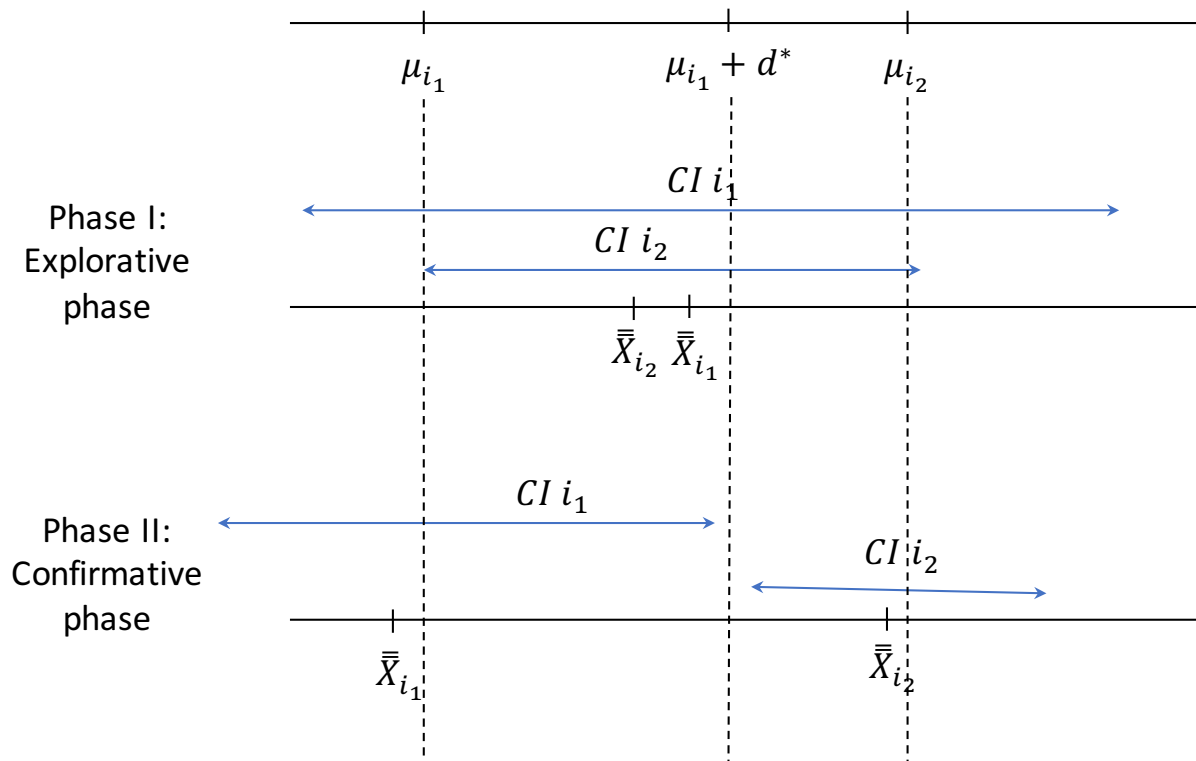
➤ Problem formulation



Comparison between design alternatives

- (on the blackboard – not in your slides)

➤ Problem formulation



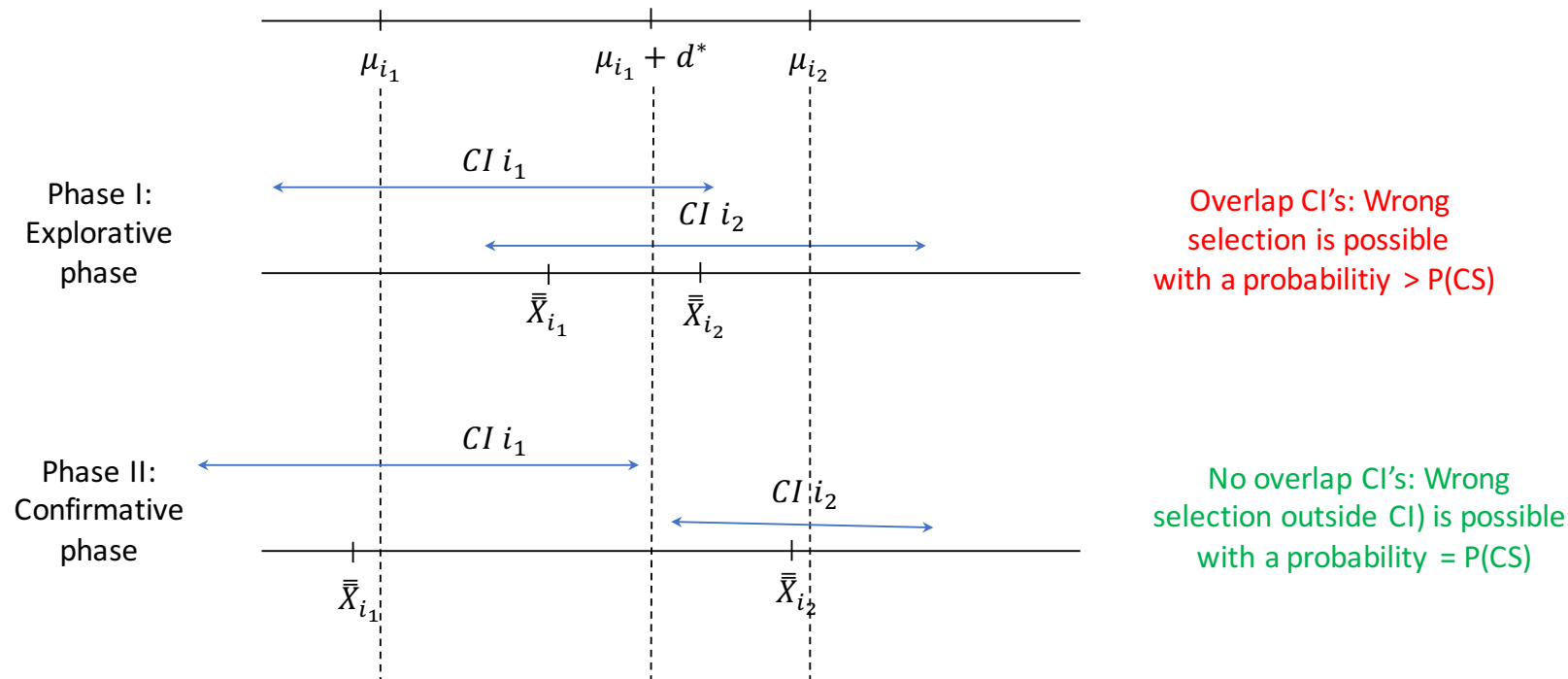
Selection of system i_2
based on sample means
⇒ Wrong selection as a result of
inaccuracy of point estimator
and too wide confidence interval

Selection of system i_1
based on sample means
⇒ Correct selection as a result of
Improved accuracy of estimators
resulting from an increased
number of samples

Comparison between design alternatives

- (on the blackboard – not in your slides)

➤ Problem formulation



Comparison between design alternatives

- Ranking and selection of design alternatives

- Example: An Inventory Problem

- Suppose that we want to compare 5 different systems
- Performance measure: Expected total cost for the first 120 months of operation (μ_i)
- Goal: Select the system with the smallest μ_i with
 - $P^* = 0.9$ to make the correct selection
 - $\mu_{i_2} - \mu_{i_1} \geq d^* = 1$ and
 - $n_0 = 20$ independent replications in the first stage of each system

=> $h_1 = 2.747$

- Results

i	$\bar{X}_i^{(1)}(20)$	$S_i^2(20)$	N_i	$\bar{X}_i^{(2)}(N_i - 20)$	W_{i1}	W_{i2}	$\bar{X}_i^{(1)}(N_i)$
1	126.48	14.52	110	124.45	0.21	0.79	124.87
2	121.92	7.96	61	121.63	0.39	0.61	121.74
3	127.16	9.45	72	126.11	0.32	0.68	126.45
4	130.71	8.25	63	132.03	0.37	0.63	131.54
5	144.07	6.20	47	144.83	0.46	0.54	144.48